

Function and description

```
boolean
Crypto_VerifyChainDER(DERCertificate[] certificates)
```

Given a list of DER-encoded certificates in [RFC 5280](#) format, starting at the end-entity (leaf) certificate, and following the chain of trust towards the root, `Crypto_VerifyChainDER()` performs all the necessary validity checks on `certificates`.

The Validity period validation for the root and optional intermediate certificates is performed against the `notBefore` timestamp of the end-entity (leaf certificate) used as value for the current time.

Mapping (Version 1.0)

```
Crypto_VerifyChain(certificates) :=
    boolean verified
```

`verified` is `TRUE` if the `Matter certificates` are verified as prescribed by [RFC 5280](#).

```
Crypto_VerifyChainDER(certificates) :=
    boolean verified
```

`verified` is `TRUE` if the `certificates` are verified as prescribed by [RFC 5280](#).

The primitives as described above verify cryptographic integrity of the certificate chains. This specification imposes a number of additional constraints on certificates discussed below in sections on [Device Attestation Certificates](#), [Node Operational Certificates](#) and [Certificate Common Conventions](#).

3.5.6. Time and date considerations for certificate path validation

The Basic Path Validation algorithm in [RFC 5280](#) mandates the consideration of the "current time" against the validity period (`notBefore`, `notAfter` fields) when validating paths. The usage of "current time" assumes that such a time is available and correct, which is a strong assumption when considering some constrained devices or devices only locally reachable on a network in the absence of infrastructure to synchronize time against a global real-time reference.

When the `Crypto_VerifyChain` primitive is used, rather than overriding the Basic Path Validation algorithm of [RFC 5280](#), Nodes SHALL consider the following definition of "current time" that accounts for the possible lack of a real time reference:

- If a Node has a current real-time clock value which is trusted according to implementation-defined means to be accurate with regard to global real-time, whether using [Time Synchronization](#) features of this specification or other means, then it SHALL use that time;
- Otherwise, the current time SHALL be set to the last-known-good UTC time.

Upon failure to validate a certificate path, where the only reason for failure is an invalid validity period of a path element, a Node MAY apply a policy of its choice to determine whether to ignore